Budapesti Műszaki és Gazdaságtudományi Egyetem
Szoftvertechnológia (VIMIAB04)

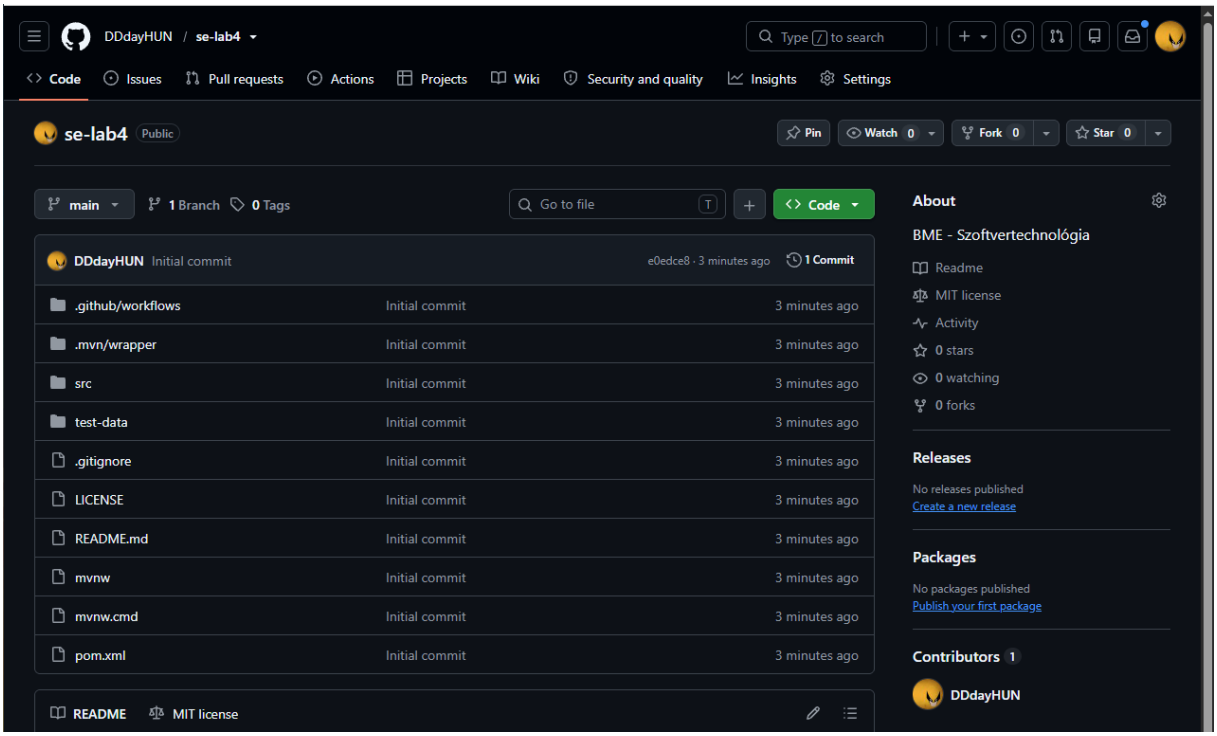


1. LAB03: CI és Statikus technikák feladatok

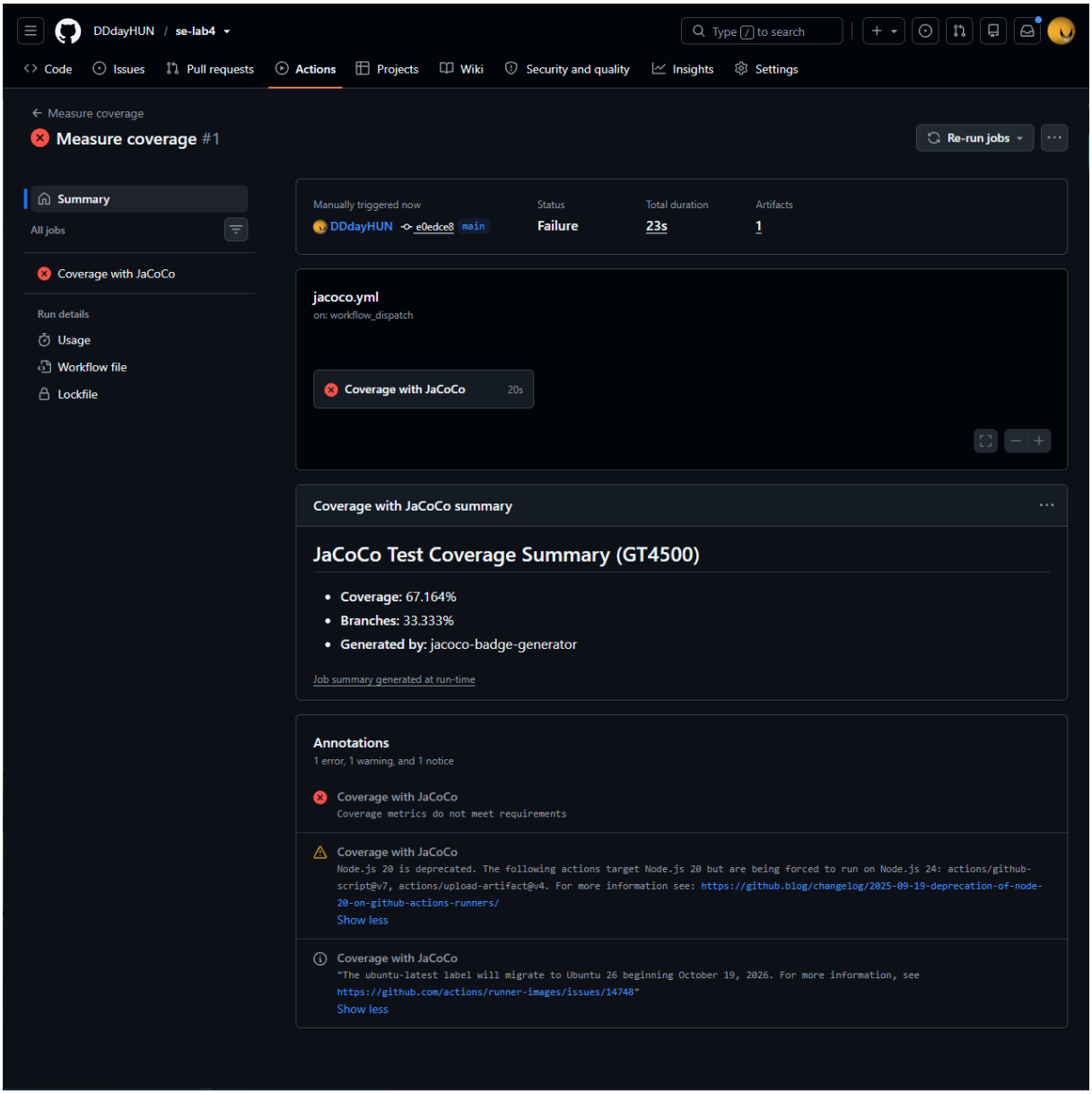
Jegyzőkönyv

Antalics Péter (R29SVR)
2026. október 27.

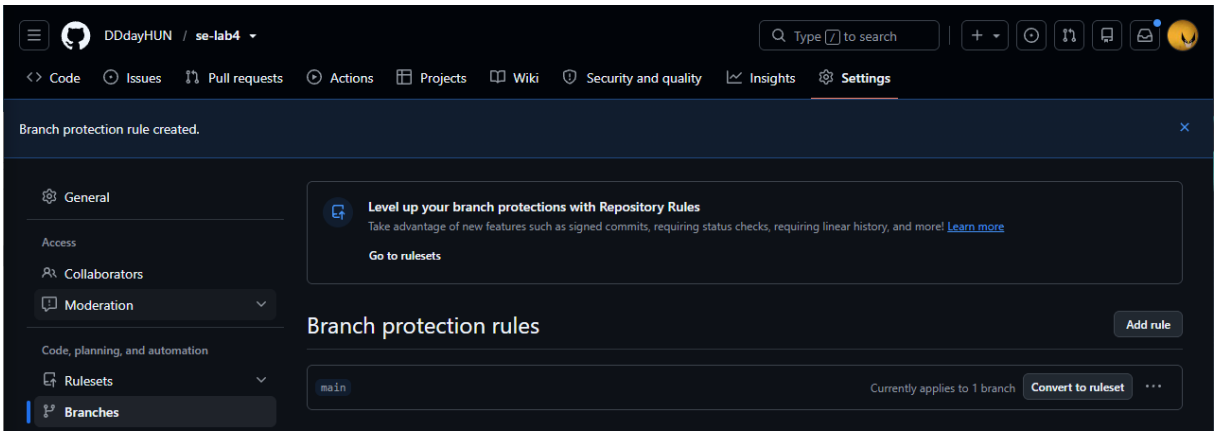
1. Feladat: GitHub tároló létrehozása



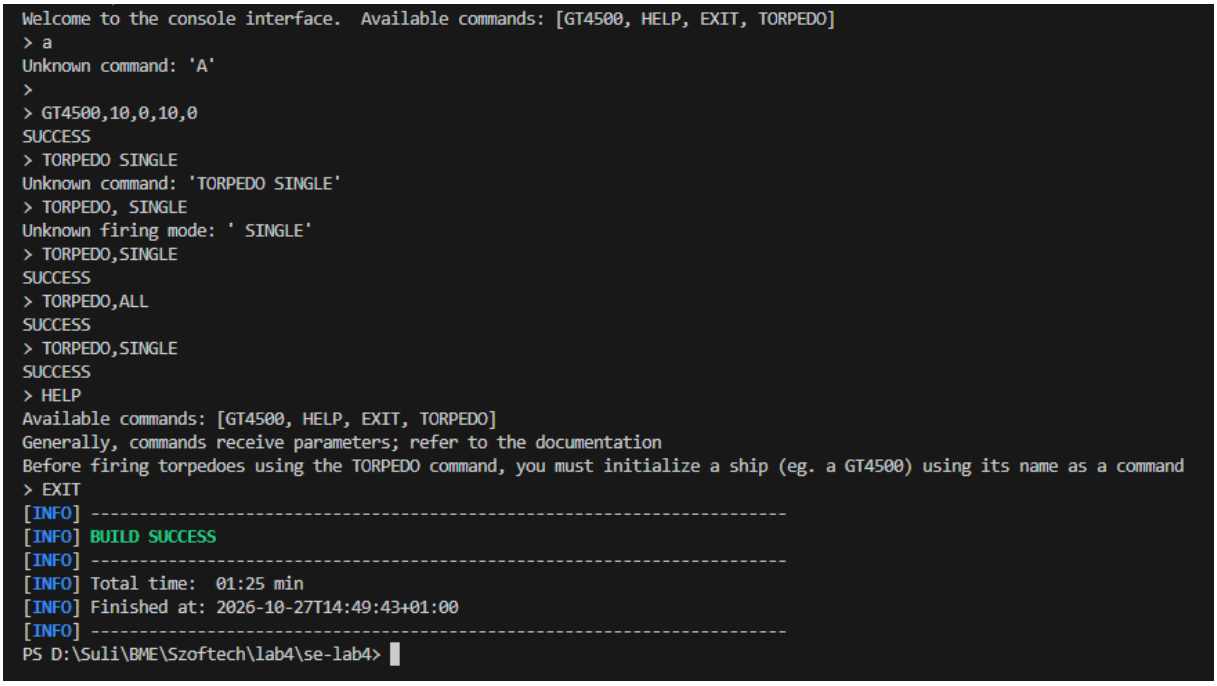
2. Feladat: Branch protection szabály konfigurálása



Az adott munkafolyamat a GT4500 fájl kódlefedettségét nézi jacoco kódtesztelővel. Az elvárt lefedettségi szint az 80%.



3. Feladat: A parancssori interfész kipróbálása




```
Kipróbált parancsok:
GT4500,10,0,10,0 -> SUCCESS
TORPEDO,SINGLE -> SUCCESS
TORPEDO,ALL -> SUCCESS
HELP -> Available commands: [GT4500, HELP, EXIT, TORPEDO]
Generally, commands receive parameters; refer to the documentation
Before firing torpedoes using the TORPEDO command, you must initialize a
ship (eg. a GT4500) using its name as a command
EXIT ->
```

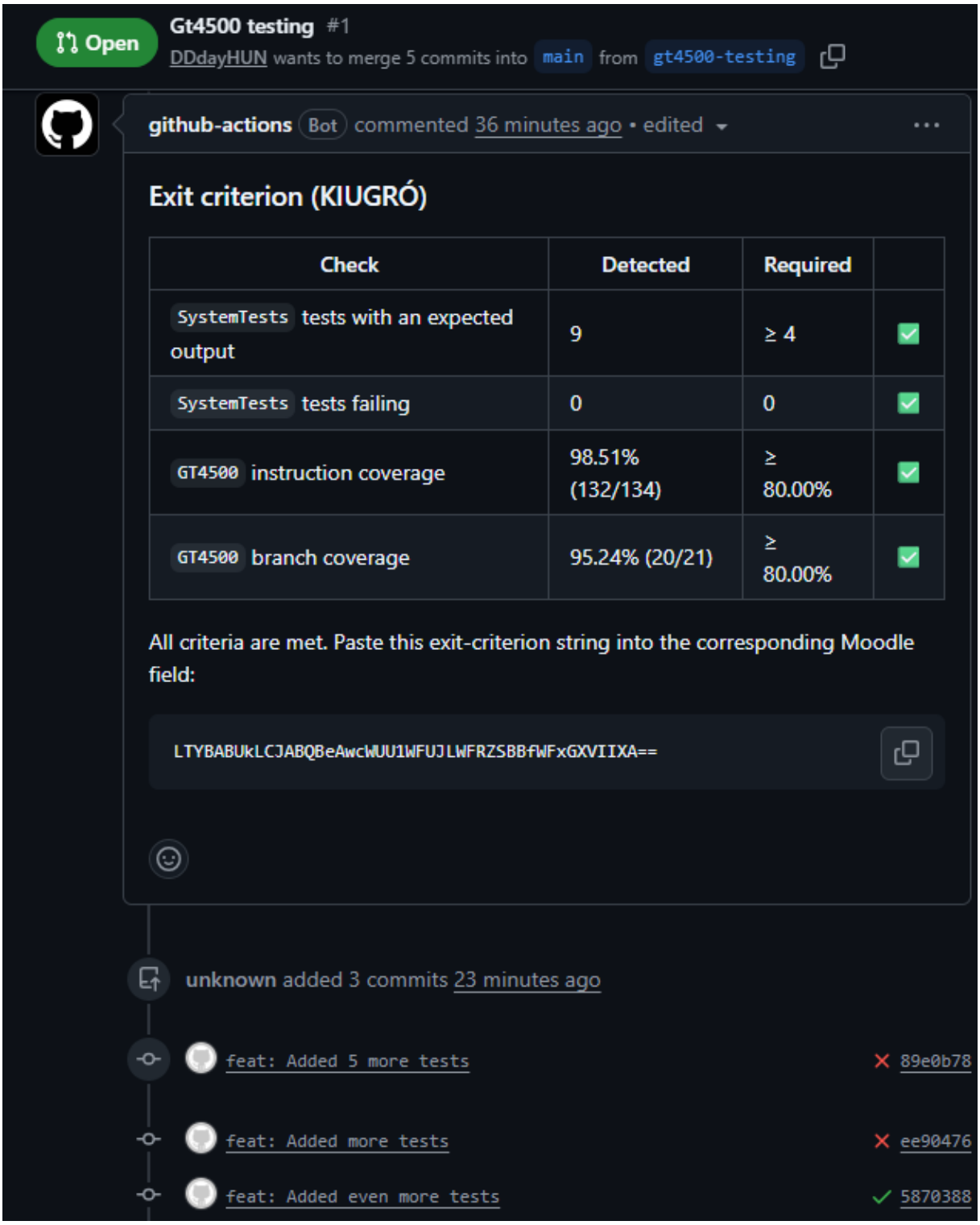
4. Feladat: Tesztttervezés

Teszt követelmény	Előfeltételek	A teszt lépései	Elvárt kimenet
A GT4500 inicializálása 0 torpedóval a primary és a secondary-ben is-is.	A GT4500 osztály példányában nincsen torpedó, se az primary, se a secondary bayben.	fireTorpedo(…) meghívása SINGLE módban.	FAIL
A GT4500 inicializálása 0 torpedóval a primary és a secondary-ben is-is.	A GT4500 osztály példányában nincsen torpedó, se az primary, se a secondary bayben.	fireTorpedo(…) meghívása ALL módban.	FAIL

A GT4500 inicializálása 1 torpedóval a primary store-ban.	A GT4500 inicializált példány 1 torpedóval a primary store-ban.	fireTorpedo(…) meghívása SINGLE módban.	SUCCESS
A GT4500 inicializálása 1 torpedóval a primary store-ban.	A GT4500 inicializált példány 1 torpedóval a primary store-ban.	fireTorpedo(…) meghívása ALL módban.	SUCCESS
A GT4500 inicializálása 1 torpedóval a secondary store-ban.	A GT4500 inicializált példány 1 torpedóval a secondary store-ban.	fireTorpedo(…) meghívása SINGLE módban.	SUCCESS
A GT4500 inicializálása 1 torpedóval a secondary store-ban.	A GT4500 inicializált példány 1 torpedóval a secondary store-ban.	fireTorpedo(…) meghívása SINGLE módban.	SUCCESS

```
PROBLEMS 3  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  SONARQUBE 2
191157@k12 MINGW64 ~/se-lab4 (gt4500-testing)
$ ./mvnw verify
[INFO] | +-- [OK] fire_Success - 0.002s
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 12, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.5.1:jar (default-jar) @ hu.bme.mit.spaceship ---
[INFO]
[INFO] --- jacoco:0.8.15:report (generate-report) @ hu.bme.mit.spaceship ---
[INFO] Loading execution data file C:\Users\191157\se-lab4\target\jacoco.exec
[INFO] Analyzed bundle 'Coverage with JaCoCo and Maven' with 6 classes
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  5.005 s
[INFO] Finished at: 2026-10-27T17:22:41+01:00
[INFO] ------------------------------------------------------------------------

191157@k12 MINGW64 ~/se-lab4 (gt4500-testing)
$
```



Open **Gt4500 testing** #1

DDdayHUN wants to merge 5 commits into `main` from `gt4500-testing`

github-actions Bot commented 36 minutes ago • edited

Exit criterion (KIUGRÓ)

Check	Detected	Required	
`SystemTests` tests with an expected output	9	≥ 4	✅
`SystemTests` tests failing	0	0	✅
`GT4500` instruction coverage	98.51% (132/134)	≥ 80.00%	✅
`GT4500` branch coverage	95.24% (20/21)	≥ 80.00%	✅

All criteria are met. Paste this exit-criterion string into the corresponding Moodle field:

```
LTYBABUkLCJABQBeAwcWUU1WFUJLWFRZSBBfWFxGXVIIXA==
```

unknown added 3 commits 23 minutes ago

- `feat: Added 5 more tests` ✗ `89e0b78`
- `feat: Added more tests` ✗ `ee90476`
- `feat: Added even more tests` ✓ `5870388`

5. Feladat: A kódlefedettség mérése




```
10.
11.   private boolean wasPrimaryFiredLast = false;
12.
13.   public GT4500() {
14.       this.primaryTorpedoStore = new TorpedoStore(10);
15.       this.secondaryTorpedoStore = new TorpedoStore(10);
16.   }
17.
18.   public GT4500(int primaryCount, double primaryFailRate, int secondaryCount,
19.           double secondaryFailRate) {
20.       this.primaryTorpedoStore = new TorpedoStore(primaryCount, primaryFailRate);
21.       this.secondaryTorpedoStore = new TorpedoStore(secondaryCount, secondaryFailRate);
22.   }
23.
24.   public boolean fireLaser(FiringMode firingMode) {
25.       // TODO not implemented yet
26.       return false;
27.   }
28.
29.   /**
30.    * Tries to fire the torpedo stores of the ship.
31.    *
32.    * @param firingMode how many torpedo bays to fire
33.    *     SINGLE: fires only one of the bays.
34.    *             - For the first time the primary store is fired.
35.    *             - To give some cooling time to the torpedo stores, torpedo stores are fired
36.    *               alternating.
37.    *             - But if the store next in line is empty, the ship tries to fire the other
38.    *               store.
39.    *             - If the fired store reports a failure, the ship does not try to fire the
40.    *               other one.
41.    *     ALL: tries to fire both of the torpedo
42.    *        stores.
43.    *
44.    * @return whether at least one torpedo was fired successfully
45.    */
46.   @Override
47.   public boolean fireTorpedo(FiringMode firingMode) {
48.
49.       boolean firingSuccess = false;
50.
51.       switch (firingMode) {
52.           case SINGLE:
53.               if (wasPrimaryFiredLast) {
54.                   // try to fire the secondary first
55.                   if (!secondaryTorpedoStore.isEmpty()) {
56.                       firingSuccess = secondaryTorpedoStore.fire(1);
57.                       wasPrimaryFiredLast = false;
58.                   } else {
59.                       // although primary was fired last time, but the secondary is empty
60.                       // thus try to fire primary again
61.                       if (!primaryTorpedoStore.isEmpty()) {
62.                           firingSuccess = primaryTorpedoStore.fire(1);
63.                           wasPrimaryFiredLast = true;
64.                       }
65.
66.                       // if both of the stores are empty, nothing can be done, return failure
67.                   }
68.               } else {
69.                   // try to fire the primary first
70.                   if (!primaryTorpedoStore.isEmpty()) {
71.                       firingSuccess = primaryTorpedoStore.fire(1);
72.                       wasPrimaryFiredLast = true;
73.                   } else {
74.                       // although secondary was fired last time, but primary is empty
75.                       // thus try to fire secondary again
76.                       if (!secondaryTorpedoStore.isEmpty()) {
77.                           firingSuccess = secondaryTorpedoStore.fire(1);
78.                           wasPrimaryFiredLast = false;
79.                       }
80.
81.                       // if both of the stores are empty, nothing can be done, return failure
82.                   }
83.               }
84.               break;
85.
86.           case ALL:
87.               // try to fire both of the torpedo stores
88.               boolean primarySuccess = false;
89.               boolean secondarySuccess = false;
90.
91.               if (!primaryTorpedoStore.isEmpty()) {
92.                   primarySuccess = primaryTorpedoStore.fire(1);
93.               }
94.               if (!secondaryTorpedoStore.isEmpty()) {
95.                   secondarySuccess = secondaryTorpedoStore.fire(1);
96.               }
97.
98.               firingSuccess = primarySuccess || secondarySuccess;
99.               break;
100.      }
101.
102.      return firingSuccess;
103.  }
```

6. Feladat: Teljes lefedettség elérése

Nem érhető el, mivel a fireLaser metódus egyeltalán nem hívható meg a jelenlegi tesztkörnyezettel.
A teljes ágazati lefedettség sem érhető el, mivel a switch statement firingMode nem tud olyan értéket felvenni, amire a switch statement defaultolna.




```
10.
11.   private boolean wasPrimaryFiredLast = false;
12.
13.   public GT4500() {
14.       this.primaryTorpedoStore = new TorpedoStore(10);
15.       this.secondaryTorpedoStore = new TorpedoStore(10);
16.   }
17.
18.   public GT4500(int primaryCount, double primaryFailRate, int secondaryCount,
19.           double secondaryFailRate) {
20.       this.primaryTorpedoStore = new TorpedoStore(primaryCount, primaryFailRate);
21.       this.secondaryTorpedoStore = new TorpedoStore(secondaryCount, secondaryFailRate);
22.   }
23.
24.   public boolean fireLaser(FiringMode firingMode) {
25.       // TODO not implemented yet
26.       return false;
27.   }
28.
29.   /**
30.    * Tries to fire the torpedo stores of the ship.
31.    *
32.    * @param firingMode how many torpedo bays to fire
33.    *     SINGLE: fires only one of the bays.
34.    *             - For the first time the primary store is fired.
35.    *             - To give some cooling time to the torpedo stores, torpedo stores are fired
36.    *               alternating.
37.    *             - But if the store next in line is empty, the ship tries to fire the other
38.    *               store.
39.    *             - If the fired store reports a failure, the ship does not try to fire the
40.    *               other one.
41.    *     ALL: tries to fire both of the torpedo
42.    *        stores.
43.    *
44.    * @return whether at least one torpedo was fired successfully
45.    */
46.   @Override
47.   public boolean fireTorpedo(FiringMode firingMode) {
48.
49.       boolean firingSuccess = false;
50.
51.       switch (firingMode) {
52.           case SINGLE:
53.               if (wasPrimaryFiredLast) {
54.                   // try to fire the secondary first
55.                   if (!secondaryTorpedoStore.isEmpty()) {
56.                       firingSuccess = secondaryTorpedoStore.fire(1);
57.                       wasPrimaryFiredLast = false;
58.                   } else {
59.                       // although primary was fired last time, but the secondary is empty
60.                       // thus try to fire primary again
61.                       if (!primaryTorpedoStore.isEmpty()) {
62.                           firingSuccess = primaryTorpedoStore.fire(1);
63.                           wasPrimaryFiredLast = true;
64.                       }
65.
66.                       // if both of the stores are empty, nothing can be done, return failure
67.                   }
68.               } else {
69.                   // try to fire the primary first
70.                   if (!primaryTorpedoStore.isEmpty()) {
71.                       firingSuccess = primaryTorpedoStore.fire(1);
72.                       wasPrimaryFiredLast = true;
73.                   } else {
74.                       // although secondary was fired last time, but primary is empty
75.                       // thus try to fire secondary again
76.                       if (!secondaryTorpedoStore.isEmpty()) {
77.                           firingSuccess = secondaryTorpedoStore.fire(1);
78.                           wasPrimaryFiredLast = false;
79.                       }
80.
81.                       // if both of the stores are empty, nothing can be done, return failure
82.                   }
83.               }
84.               break;
85.
86.           case ALL:
87.               // try to fire both of the torpedo stores
88.               boolean primarySuccess = false;
89.               boolean secondarySuccess = false;
90.
91.               if (!primaryTorpedoStore.isEmpty()) {
92.                   primarySuccess = primaryTorpedoStore.fire(1);
93.               }
94.               if (!secondaryTorpedoStore.isEmpty()) {
95.                   secondarySuccess = secondaryTorpedoStore.fire(1);
96.               }
97.
98.               firingSuccess = primarySuccess || secondarySuccess;
99.               break;
100.      }
101.
102.      return firingSuccess;
103.  }
```

7. Feladat: Egységteszt készítése

```
191157@k12 MINGW64 ~/se-lab4 (gt4500-testing)
$ ./mvnw verify
[INFO] | +-- [OK] runCommandsFromFile_Success(Path, Path)[4] test-data\input-3.txt, test-data\output-3.txt - 0.003s
[INFO] | +-- [OK] runCommandsFromFile_Success(Path, Path)[5] test-data\input-4.txt, test-data\output-4.txt - 0.002s
[INFO] | +-- [OK] runCommandsFromFile_Success(Path, Path)[6] test-data\input-5.txt, test-data\output-5.txt - 0.003s
[INFO] | +-- [OK] runCommandsFromFile_Success(Path, Path)[7] test-data\input-6.txt, test-data\output-6.txt - 0.003s
[INFO] | +-- [OK] runCommandsFromFile_Success(Path, Path)[8] test-data\input-7.txt, test-data\output-7.txt - 0.003s
[INFO] | +-- [OK] runCommandsFromFile_Success(Path, Path)[9] test-data\input-8.txt, test-data\output-8.txt - 0.004s
[INFO] |
[INFO] +-- hu.bme.mit.spaceship.TorpedoStoreTest
[INFO] | +-- [OK] fireFailedNoTorpedoesInStore - 0.005s
[INFO] | +-- [OK] failure - 0.002s
[INFO] | +-- [OK] fire_Success - 0.002s
[INFO] | +-- [OK] getCount - 0.002s
[INFO] | +-- [OK] isEmpty - 0.002s
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 16, Failures: 0, Errors: 0, Skipped: 0
```

```java
class TorpedoStoreTest {

    @Test
    void fire_Success() {
        // Arrange
        TorpedoStore store = new TorpedoStore(1);

        // Act
        boolean result = store.fire(1);

        // Assert
        assertEquals(true, result);
    }

    @Test
    void fireFailedNoTorpedoesInStore() {
        // Arrange
        TorpedoStore store = new TorpedoStore(0);

        // Assert
        assertThrows(IllegalArgumentException.class, () -> store.fire(2));
    }

    @Test
    void isEmpty() {
        // Arrange
        TorpedoStore store1 = new TorpedoStore(0);
        TorpedoStore store2 = new TorpedoStore(1);

        // Assert
        assertEquals(true, store1.isEmpty());
        assertEquals(false, store2.isEmpty());
    }
```

```
    @Test
    void getCount() {
        // Arrange
        TorpedoStore store1 = new TorpedoStore(0);
        TorpedoStore store2 = new TorpedoStore(5);

        // Assert
        assertEquals(0, store1.getTorpedoCount());
        assertEquals(5, store2.getTorpedoCount());
    }

    @Test
    void failure() {
        // Arrange
        TorpedoStore store = new TorpedoStore(1, 1.0);

        // Assert
        assertEquals(false, store.fire(1));
    }
}
```